

Part A

Q1

```

DECLARE i = INTEGER.
INPUT N
i ← 0
FOR i ← N to 1 STEP -1
    OUTPUT i
END FOR.

```

Q2

```

INPUT Year.
IF Year MOD 400 = 0 THEN
    OUTPUT "It is a leap Year".
ELSE IF Year MOD 100 = 0 THEN
    OUTPUT "it is Not a leap year".
ELSE IF Year MOD 4 = 0 THEN
    OUTPUT "It is a leap Year"
ELSE
    OUTPUT "It is NOT a leap Year"

```

Q3

DECLAR X: STRING, DECLARE Y: INTEGER

INPUT Number

X ← NUM_TO_STR(Number)

~~Y ← length~~

Y ← LENGTH(X)

IF Y = 1 THEN

OUTPUT "It is a single Digit Number"

ELSE IF Y = 2 THEN

OUTPUT "It is a Double Digit Number"

ELSE IF Y = 3 THEN

OUTPUT "It is a Triple Digit Number"

ELSE

OUTPUT "It is a More than 3 digits

Number"

ENDIF

```

Q4. DECLARE Index, Multiple, i: INTEGER.
      Index  $\leftarrow$  1, i  $\leftarrow$  0, Multiple  $\leftarrow$  0.
      WHILE Index  $\leq$  10
      IF Index MOD 3  $\neq$  0 THEN
      FOR. i  $\leftarrow$  1 to 10.
      Multiple  $\leftarrow$  Index * i
      OUTPUT Index, "*", i, "=", Multiple.
      Next
      GNDIF
      Index  $\leftarrow$  Index + 1.
      ENDWHILE
  
```

```

Q5 DECLARE Sum, i: INTEGER.
   DECLARE NStr: STRING.
   INPUT NUM, NUM
   NStr ← NUM_TO_STR(NUM)
   Sum ← 0, i ← 0.
   FOR i ← 1 to LENGTH(NStr).
       Sum ← Sum + MID(NStr, i, 1).
   NEXT

```

① OUTPUT Sum